

32 – Sync Contract

The contract between a till and central: what moves in each direction, in what unit, in what order, how it is acknowledged, how versions and gaps are handled, and what happens on every failure that matters. This is the one interface that is expensive to change once terminals are in the field, so it is specified before the till is designed.

Field	Value
Document	32 – Sync Contract (design artefact; Stage 1/2 boundary; owned by the till track with the kernel's edge sync gateway)
Version	v0.1 – draft for design review
Date	7 September 2026
Status	Draft. Batch sizes, retention windows and the push mechanism are decision requests (Section 12); the conformance suite (Section 11) is the acceptance criterion.
Inputs	ADR-10 (one writer), ADR-11 (numbering), ADR-24, ADR-31 (shared engine), ADR-34 (snapshot distribution); NFR-PRF-030/040, NFR-OFF-010/030, NFR-AVL-020/050; H-01 to H-04, Q-06; doc 17 v0.2.1 sections 8 and 9; doc 18 parts C and D; doc 19 section 6; doc 14 section 6; doc 20 M6; doc 31.
Audience	Till track, kernel developers, DevOps, test engineers. Section 1 for reviewers.

The contract in one paragraph. A till writes its own facts into a local outbox with a dense sequence number. It uploads them in order, one batch in flight at a time, and central acknowledges the highest sequence it has durably applied. Central never edits, re-numbers or merges what a till sends; a fact it cannot accept is quarantined and flagged, never lost and never allowed to block what follows. In the other direction, central publishes versioned snapshots of reference data per shop; a till downloads the delta since its version and applies it atomically. No record has two writers, so there is nothing to reconcile.

1. Principles the contract enforces

#	Principle	Consequence in the protocol
S1	One writer per record type (ADR-10)	Channel 1 carries only till-owned records; channel 2 carries only central-owned records. Nothing appears in both.
S2	Sequence orders events, never clocks	Every uploaded event carries a dense per-device sequence; central applies in that order; clocks are recorded for reporting and corrected for drift, never used to order.
S3	Exactly-once effect over at-least-once delivery	The till may send a batch twice; central applies each event id once; the acknowledgement is derived from database state, so any instance can answer.
S4	Never reject a fact, quarantine a malformed one	A sale that happened is applied even if it breaks a business rule (flagged); a payload that cannot be parsed is stored raw, the sequence advances, an alert is raised.
S5	Atomic reference data	A snapshot version is applied whole or not at all; a bill is never computed against a half-applied price list.
S6	Resumable, small, compressible	Batches are bounded; downloads are resumable byte ranges; everything is gzip-compressed; the weakest link (about 2 Mbps down, 0.5 up) is the design case.
S7	Stateless central	No instance memory participates; the device cursor row in the database is the only state (doc 14 section 1.3).
S8	Nothing is deleted at the edge until it is safe	Acknowledged outbox rows are retained for a retention window so central can ask for them again after a point-in-time restore.

2. Identities and state

Item	Where held	Meaning
Device credential	Terminal keystore; central device table	Client certificate (or secret) bound to the hardware serial; authenticates every call (doc 19 section 2.1)
Till position	Central; cached on device	The logical lane; the till-position numbering series follows it
Primary till flag	Central location.primary_till_position_id; cached	Whether this device holds the location-scoped series (GRN, count, write-off draft, transfer)
Device sequence (device_seq)	Device outbox; central cursor	Dense, monotonic, per device, across all record kinds; starts at 1 at enrolment; never reset except by the audited recovery procedure (Section 8)
Sync cursor	Central table device_sync_cursor: device_id, last_applied_seq, last_batch_id, last_ack_at, snapshot_version_reported	The only server-side sync state; row-locked during ingestion
Snapshot version	Central per-location version counter; device local	Monotonic integer per location; the device reports the version it trades on with every upload
Operator context	Per event	The authenticated operator (user id) and role snapshot version at the time

3. Channel 1: event upload (till to central)

3.1 What is uploaded

Every till-owned fact is an event in the doc 19 envelope. The till emits four families; each is applied by a kernel or module handler at central.

Family	Event types	Payload	Applied by
Document bundle	receipt.issued/voided/refunded, grn.captured/confirmed, count.recorded, writeoff.requested (draft), repack.executed, transfer.issued (intra-entity at shop)	The complete document: header, lines, extension table rows, links, attachment references with hashes, and the DRAFT→ISSUED state row; content_hash	Kernel document base inserts; owning module's handler (M6, M4, M5) validates and publishes to the backbone
Stock movement	stock.moved	Movement rows for one document, cited by document id	M5 ledger; balances recomputed
Audit	audit.* (every audited action on the till, including mode switches, negative-stock acknowledgements, MRP picker use, drawer-open-no-sale)	Audit row fields (doc 18 part D)	Kernel audit store (insert only)
Till lifecycle	till_session.opened/closed, cash.dropped, till.heartbeat, sync.local_anomaly	Session and health facts	M6 central side; fleet and freshness views

A document bundle is one event so that its application is atomic: header, lines, movements and the state row commit together or not at all. Stock movements for a receipt travel inside the receipt bundle; the separate stock.moved family exists for movements that have no bundle of their own (a correction applied by a later procedure).

3.2 The batch

Aspect	Design
Unit	A batch is a contiguous run of events in device_seq order. Up to 500 events or 2 MB compressed, whichever first (DR-1). A batch never splits a document bundle.
In flight	Exactly one batch in flight per device. The till does not send the next batch until it has an acknowledgement or a timeout; central rejects a second concurrent batch for the same device with 409 and the in-flight batch id.
Request	POST /sync/v1/devices/{device_id}/batches with device authentication; gzip body: batch_id (UUIDv7), first_seq, last_seq, app_version, snapshot_version_in_use, device_clock, device_uptime, events[].
Response	200 with ack (Section 3.4); 409 for sequence mismatch or concurrent batch, carrying expected_seq; 429 with retry_after under backpressure; 401/403 for credential or suspension; 5xx retried with backoff.
Timeouts and retry	Client timeout 60 s on the weakest link; exponential backoff with jitter from 5 s to 10 min; a lost response is not an error: the retry carries the same batch_id and central answers idempotently.
Priority under backlog	Batches are sent in sequence; there is no reordering by kind. To keep receipts flowing first after an outage the till emits audit rows for a document immediately after its bundle, so sequence order already interleaves them tightly.

3.3 Central ingestion

Step	Action	Rule
1	Authenticate the device; check status ACTIVE; check app_version against the floor (doc 31)	Suspended device: 403 and a signed revoke instruction; below floor after grace: 426 with reason
2	SELECT device_sync_cursor FOR UPDATE	One ingestion per device at a time on any instance (S7)
3	Compare first_seq with last_applied_seq	first_seq = last+1: proceed. first_seq ≤ last: the batch (or its prefix) is a replay; answer from state without re-applying. first_seq > last+1: gap; 409 with expected_seq (Section 7)
4	Apply events in order in chunks of 50 per transaction	Each event: skip if event_id already in the inbox; else validate envelope and schema; verify content_hash for bundles; insert; publish to the outbox; record in the inbox
5	Business validation with the shared engine (ADR-31)	Re-resolve price and ceilings against the snapshot version the till reported; any difference or ceiling breach is recorded as a flag on the line and an ALERT audit event; the event is still applied (S4)
6	Malformed event	Store raw in sync_quarantine with the reason; advance the cursor; raise SYNC_ANOMALY ALERT; continue (S4)

Step	Action	Rule
7	Update the cursor to the highest applied seq; commit; release the lock	Cursor advances only on commit; a crash leaves the cursor at the last committed chunk
8	Return the ack	Section 3.4

3.4 Acknowledgement

Aspect	Design
Content	batch_id; last_applied_seq (authoritative); per-event outcomes: APPLIED, DUPLICATE, FLAGGED (with codes), QUARANTINED (with reason); server_time; clock_offset_ms; current snapshot_version for the location; pending instructions (revoke, floor notice, re-request from seq, series holder change).
Meaning	Everything at or below last_applied_seq is durably at central. The till marks those outbox rows acknowledged and retains them for the retention window (S8, DR-3: 7 days), then purges.
Partial success	If ingestion failed part-way, last_applied_seq is the last committed chunk; the till resends from last_applied_seq + 1 with a new batch_id. Already-applied events in the overlap are answered DUPLICATE.
Lost ack	The till resends the same batch_id; central finds first_seq ≤ cursor and answers from state; no double application (S3).

4. Attachments

Aspect	Design
Path	Photographs never travel inside a batch. The till requests a pre-signed PUT for each attachment (object key derived from document id and attachment id), uploads directly to object storage, resumable in byte ranges, then includes the key and content hash in the document bundle or in a later attachment.completed event.
Ordering	A document may be applied before its attachments finish uploading; the document carries the expected keys and hashes; central marks attachments PENDING until the object exists and the hash matches, then COMPLETE. A write-off approval requires all its attachments COMPLETE.
Limits	Images are downscaled on the device to at most 1600 px on the long edge and about 300 KB; a bundle references at most 10 attachments.

5. Channel 2: reference-data distribution (central to till)

5.1 Scope of a shop's snapshot

Table set	Content for the location	Source module
Catalogue	SKUs in the shop's assortment: on the MPCS retail list, or with a lot at the location, or local to the owning entity; with units, conversions, barcodes, tags, tax categories, thumbnail keys	M2
Pricing	The MPCS retail price list lines; active and scheduled discount rules; control prices in force or scheduled; multi-MRP policies; rounding rule; tax rates	M3
Lots	Batches with positive on-hand at the location: batch id, SKU, expiry, printed MRP, FEFO rank, quantity as last known	M5 (batch identity from M2)
Customers	Tagged and credit customers of the MPCS with balance, limit, offline cap	M7
Operators and permissions	Operators assigned to the location: user id, display name, PIN hash, resolved offline permissions with catalogue version; signed	M1 via doc 19 section 3.3
Message catalogue and configuration	Translations for the interface; configuration items for the location (idle locks, quiet hours, offline account cap, thresholds); series assignments and primary-till flag	Kernel

5.2 Versions and deltas

Aspect	Design
Version	Central keeps one monotonic snapshot version per location. Any publication that changes the shop's snapshot (price list published, rule activated, control price entered, SKU shared, lot changed by a GRN elsewhere in the entity, operator added) records a change-log row and bumps the version.
Delta request	GET /sync/v1/locations/{location_id}/snapshot?since=N returns a delta from N+1 to current: per table, upserted rows and tombstones, with the new version. If N is older than the change-log retention (DR-4: 30 days) or the device has no version, the response points to a full snapshot.
Full snapshot	A pre-built compressed file in object storage per location, rebuilt when the delta would exceed a threshold or nightly, with a manifest (tables, row counts, hashes, version) and a pre-signed resumable download. Target under 20 MB with thumbnails.
Atomic apply	The device loads a delta or full snapshot into staging tables, verifies hashes, then swaps into the live cache and records the version in one local transaction. Selling continues on the old version until the swap; the swap waits for the current sale to complete.
apply_from	Rows with an apply_from business date later than the shop's current business date are held in pending and activated at the shop's day-open, so every till in the federation switches prices on the same trading day regardless of when it downloaded.
Cadence	Version check on every heartbeat (5 min) when online; download at most every 15 minutes (NFR-PRF-040) unless central marks the change urgent (control price entered, permission revoked), in which case the next heartbeat downloads immediately. An optional push wake-up (DR-2) shortens this; the design does not depend on it.
Consistency with uploads	Every upload reports snapshot_version_in_use; central re-validates prices against that version, so a legitimate sale made on a slightly stale list is recognised as such rather than flagged as an error.
Staleness warning	A till whose snapshot is older than a configured age (default 3 days) warns the supervisor that ceilings and prices may be stale; it keeps selling.

6. Heartbeat and health

Aspect	Design
Cadence	Every 5 minutes when online; on app start; after any batch.
Content	app_version, snapshot_version, pending_event_count, oldest_pending_seq_age, last_applied_seq as known, battery, storage free, peripheral self-test summary, device clock and uptime, open session flag.
Central use	device.last_seen_at and app_version; per-location freshness stamp = time of the last applied event from the location's devices; fleet console; rollout health (doc 31); staleness detection.
Clock	Central returns the offset; the till stores it and stamps occurred_local plus the offset on later events; large drift (over 10 minutes) raises a REVIEW audit event but never blocks.

7. Gap and anomaly handling

Situation	Detection	Handling
Batch starts above the cursor	$\text{first_seq} > \text{last_applied_seq} + 1$	409 with expected_seq. The till must resend from expected_seq; it can, because unacknowledged rows are never purged. Repeated 409 for the same device raises SYNC_ANOMALY.
Batch starts at or below the cursor	$\text{first_seq} \leq \text{last_applied_seq}$	Replay: answered from state; events above the cursor within the batch are applied normally.
Duplicate event id with a different sequence	Inbox has event_id at another seq	Quarantine the later copy; ALERT; advance cursor.
Hash mismatch on a bundle	Recomputed content_hash differs	Quarantine; ALERT; advance cursor. The document is not applied until the till resends a correct copy through the recovery procedure.
Document number gap within a series	Kernel gap check over (series, doc_number) after ingestion	SYNC_ANOMALY REVIEW; a gap can be legitimate only if the device's quarantine holds the missing number; otherwise investigate.
Till outbox lost (local database corruption or replacement without transfer)	Device reports device_seq below the cursor, or cannot produce expected_seq	Never silent. The recovery procedure (Section 8) creates an audited sequence-reset record; missing events remain a documented gap; ALERT to the MPCS administrator.
Central restored from backup	Central cursor below the till's acknowledged seq	Central asks for re-send from its cursor via the ack's pending instructions; the till still holds acknowledged rows within the retention window (S8), so the data is recovered. This is why retention is not zero.
Clock drift	Offset over threshold	REVIEW event; reporting uses corrected times.
Malformed payload	Schema validation fails	Quarantine, advance, ALERT (S4).

8. Registration, series and counter transfer

Procedure	Steps	Audit
Device enrolment	Provisioned as a managed kiosk (doc 31); first start calls POST /sync/v1/devices/{id}/enrol with the credential; central assigns the till position (from staging) and returns the device_seq starting value (1), series assignments (till-position series; location series if primary), and the initial snapshot pointer	device.enrolled, series.registered
Series assignment	Central creates numbering_series rows for the position; the device persists counters from the last synchronised next_number (1 for new series)	series.registered
Counter transfer (replacement device)	Precondition: the old device's outbox is fully acknowledged (or its loss is recorded). The administrator reassigns the position to the new device; central returns next_number per series computed from the highest applied doc_number; the new device starts there. If the old device had unacknowledged issued documents, the transfer records the numbers reserved but not received as a documented gap	device.position_changed, series.holder_changed
Primary till change	Location-scoped series move with the primary flag using the same transfer rule	series.holder_changed
Sequence reset (recovery)	Only by an MPCS administrator with a reason code, on a device that cannot produce expected_seq; central records the gap range, sets the cursor to the device's new start, raises ALERT	sync.sequence_reset

9. Security

Aspect	Design
Transport	TLS 1.2 or later to the edge; device client certificate or bearer token from the device credential; no anonymous endpoint.
Integrity	content_hash on every bundle; signed permission snapshot; snapshot manifest hashes; the ack is over the same TLS channel, so it needs no separate signature.
Replay	batch_id plus sequence semantics make replays harmless; event ids are UUIDv7 and unique.
Suspension	A suspended device receives 403 and a signed revoke; the application wipes caches and locks; unacknowledged outbox rows are preserved for recovery by an administrator.
Data minimisation	Customer snapshot carries name, phone, balance, limit only; PIN hashes are Argon2id; no card data exists.
Rate limiting	Per device: batches per minute and bytes per hour; global: ingest capacity; 429 with retry_after; prioritised fairness so one device cannot starve others.

10. Volumes and limits (planning basis, doc 14 section 6)

Quantity	Value	Note
Events per receipt	About 3 (bundle, one or two audit rows)	Movements travel inside the bundle
Bundle size	About 2–4 KB compressed	Five lines, tenders, state row
Steady upload per till	A few KB per minute	Negligible on a 0.5 Mbps uplink
Two-day backlog per till	About 400 receipts at 200 a day; about 1–2 MB; two to four batches	Drains in under a minute on the weakest link
Fleet reconnection burst	Thousands of tills, two days each; central target 3,500 events/s	Rate limiting spreads it; projections lag with freshness stamps
Full snapshot	Under 20 MB with thumbnails; under 5 MB without	Resumable download; nightly rebuild
Delta	Typically under 100 KB	Price list day: up to a few MB for a large assortment
Acknowledged outbox retention	7 days (DR-3)	Bounded by storage: about 10 MB per till

11. Conformance suite (acceptance criterion for both sides)

A simulator on each side is part of the deliverable: a central simulator the till is tested against, and a till simulator central is load-tested with (spike 5). The suite is the definition of done for this contract.

Test	Asserts
Lost acknowledgement	Same batch resent; every event applied once; cursor correct

Test	Asserts
Crash mid-batch at central	Cursor at last committed chunk; resend from cursor+1; no duplicates, no loss
Power loss on the till during issuance	Either the document and its number both exist or neither; outbox sequence dense (spike 1)
Gap	Batch above cursor rejected with expected_seq; till recovers from retained rows
Malformed event	Quarantined, cursor advanced, later events applied, ALERT raised
Business-rule breach	Event applied, line flagged, ALERT raised; nothing rejected
Concurrent batches for one device	Second rejected with 409; first completes
Two instances	Batches from one device alternate between instances; ordering and idempotency hold
Snapshot atomicity	Kill the till mid-apply; it trades on the old version; re-apply completes
apply_from	Prices dated tomorrow do not sell today; activate at day-open
Central point-in-time restore	Central re-requests from a lower cursor; till resends retained rows; state converges
Counter transfer	New device continues the series without gap or duplicate; documented gap when the old device lost data
Burst	Simulated fleet backlog absorbed within the target; fairness across devices

12. Decision requests

#	Point	Proposed	Decider
DR-1	Batch limits	500 events or 2 MB compressed	Till track and DevOps after spike 5
DR-2	Push wake-up mechanism	None required; optional MQTT-class push added if heartbeat latency proves insufficient for urgent control-price changes	Engineering lead
DR-3	Acknowledged outbox retention on the till	7 days	DevOps (restore drill informs)
DR-4	Change-log retention for deltas	30 days; older devices take a full snapshot	DevOps
DR-5	Chunk size per ingestion transaction	50 events	Kernel developers after load test
DR-6	Staleness warning age	3 days	Federation

13. Traceability

Section	Requirements and decisions
1–3	ADR-10, ADR-31; NFR-OFF-010/030; NFR-PRF-030; H-01, H-04; doc 19 6.1–6.3; doc 18 5.5
4	D-08, F-05 (photographic evidence); doc 18 document_attachment
5	ADR-34; NFR-PRF-040; G-08 (apply on propagation); doc 19 3.3 (signed permission snapshot); doc 20 M2 thumbnails
6	O-03 (freshness stamps); doc 31 health gates
7–8	ADR-11; FR-BIL-090 (counter transfer); J-05; N-05 (sync anomalies audited)
9	NFR-SEC set; ADR-27; doc 19 2.1
10–11	ADR-22; doc 14 sections 6 and 8 (spikes 1, 5)

End of document. Draft for design review; message schemas are published as an OpenAPI slice and JSON schemas alongside this document once DR-1 to DR-5 are settled.